CLASS 8 MATHEMATICS SOLUTIONS (English Medium)
SELF ASSESSMENT TERM 1 MODEL PAPER – SET 1 (2026 – 2027)
SECTION A – MULTIPLE CHOICE QUESTIONS (20 × 1 = 20 Marks)

1. Given $125^2 = 15625$, what is the value of 126^2? **[D]**

A) 15625 + 126 B) 15625 + 26^2 C) 15625 + 253 D) 15625 + 251

$(n + 1)^2 = n^2 + n + (n + 1)$
Take n = 125: $126^2 = 125^2 + 125 + 126$
$126^2 = 15625 + 251$
Final Answer: D) 15625 + 251 **(1 M)**

2. A sheet of paper has thickness v. If it is folded in half 10 times, the thickness of the folded paper is **[D]**

A) 10 × v B) 10 + v C) 2 × 10 × v D) $2^{10} \times v$

Each fold doubles the thickness.
After 1 fold: 2v; after 2 folds: $2 \times 2v = 2^2v$
After 10 folds: $2^{10} \times v$
Final Answer: D) $2^{10} \times v$ **(1 M)**

3. The base of the Mesopotamian (Babylonian) number system is **[C]**

A) 10 B) 20 C) 60 D) 100

The Mesopotamians used a sexagesimal (base-60) place value system.
(We still use it today: 1 hour = 60 minutes, 1 minute = 60 seconds.)
Final Answer: C) 60 **(1 M)**

4. The sum of the four interior angles of a quadrilateral is **[C]**

A) 180° B) 270° C) 360° D) 540°

A diagonal divides a quadrilateral into two triangles.
Angle sum = 180° + 180° = 360°
Final Answer: C) 360° **(1 M)**

5. Which of the following numbers is divisible by 9? **[B]**

A) 123 B) 405 C) 8888 D) 93547

A number is divisible by 9 if the sum of its digits is divisible by 9.
123 → 1+2+3 = 6; 405 → 4+0+5 = 9; 8888 → 32; 93547 → 28
Only 405 has digit sum 9.
Final Answer: B) 405 **(1 M)**

6. Using the identity $(a + b)(a - b) = a^2 - b^2$, the value of 45 × 55 is **[C]**

A) 2375 B) 2425 C) 2475 D) 2500

$45 \times 55 = (50 - 5)(50 + 5)$
$= 50^2 - 5^2$
$= 2500 - 25 = 2475$
Final Answer: C) 2475 **(1 M)**

7. Anagh mixes 600 mL of orange juice with 900 mL of apple juice. The ratio of orange juice to apple juice in its simplest form is **[A]**

A) 2 : 3 B) 3 : 2 C) 6 : 9 D) 3 : 4

Ratio = 600 : 900
HCF of 600 and 900 = 300
Dividing both terms by 300 → 2 : 3
Final Answer: A) 2 : 3 **(1 M)**

8. The number of natural numbers that lie between 16^2 and 17^2 is **[C]**

A) 30 B) 31 C) 32 D) 33

$16^2 = 256$ and $17^2 = 289$
Numbers between them: 257, 258, …, 288
Count = 288 − 257 + 1 = 32 (also 2 × 16 = 32)
Final Answer: C) 32 **(1 M)**

9. A digital locker has an alpha-numeric passcode (digits 0–9 and letters A–Z) of length 5. The number of possible codes is **[C]**

A) 36 × 5 B) 5^{36} C) 36^5 D) 10^5

Each place can be filled in 10 + 26 = 36 ways.
There are 5 places: $36 \times 36 \times 36 \times 36 \times 36 = 36^5$
Final Answer: C) 36^5 **(1 M)**

10. The landmark numbers of the Hindu number system are 1, 10, 100, … . The fourth landmark number is **[C]**
A) 40 B) 400 C) 1000 D) 10000
Landmark numbers: 1st = 1, 2nd = 10, 3rd = 100
4th = 10 × 10 × 10 = 1000
Final Answer: C) 1000 **(1 M)**

11. In which quadrilateral are the diagonals always equal in length and bisect each other? **[C]**
A) Parallelogram B) Rhombus C) Rectangle D) Kite
Parallelogram: diagonals bisect each other but are not always equal.
Rhombus: diagonals bisect each other at 90° but are not equal.
Kite: diagonals do not bisect each other.
Rectangle: diagonals are equal and bisect each other.
Final Answer: C) Rectangle **(1 M)**

12. A number that leaves remainder 3 when divided by 5 is of the form **[D]**
A) 3k + 5 B) 3k − 5 C) 5k − 2 D) 5k + 3
Dividend = divisor × quotient + remainder
= 5 × k + 3 = 5k + 3
Final Answer: D) 5k + 3 **(1 M)**

13. If both numbers a and b of the product a × b are increased by 1, the product increases by **[B]**
A) a + b B) a + b + 1 C) ab + 1 D) 1
New product = (a + 1)(b + 1) = ab + a + b + 1
Increase = (ab + a + b + 1) − ab = a + b + 1
Final Answer: B) a + b + 1 **(1 M)**

14. If 6 : 10 :: 18 : ?, then the missing term is **[C]**
A) 20 B) 25 C) 30 D) 36
First term changes from 6 to 18, i.e. 18 ÷ 6 = 3 times.
Second term must also become 3 times: 10 × 3 = 30
Final Answer: C) 30 **(1 M)**

15. In the Mayan number system, the third landmark number is **[B]**
A) 60 B) 360 C) 400 D) 3600
Mayan landmark numbers: 1, 20, 18 × 20 = 360, …
The third landmark number is 360.
Final Answer: B) 360 **(1 M)**

16. Which quadrilateral is both a kite and a parallelogram? **[C]**
A) Rectangle B) Trapezium C) Rhombus D) None of these
A rhombus has all four sides equal, so it has two pairs of equal adjacent sides (kite).
Its opposite sides are parallel (parallelogram).
A rectangle is not a kite in general; a trapezium is not a parallelogram.
Final Answer: C) Rhombus **(1 M)**

17. Which of the following numbers is divisible by 11? **[D]**
A) 158 B) 841 C) 481 D) 90904
Rule: (sum of digits in odd places) − (sum of digits in even places) must be 0 or a multiple of 11.
158 → (8 + 1) − 5 = 4; 841 → (1 + 8) − 4 = 5; 481 → (1 + 4) − 8 = −3
90904 → (4 + 9 + 9) − (0 + 0) = 22 = 11 × 2 ✓
Final Answer: D) 90904 **(1 M)**

18. The expansion of $(a - b)^2$ is **[C]**
A) $a^2 - b^2$ B) $a^2 + 2ab + b^2$ C) $a^2 - 2ab + b^2$ D) $a^2 - ab + b^2$
$(a - b)^2 = (a - b)(a - b)$
$= a^2 - ab - ba + b^2$
$= a^2 - 2ab + b^2$
Final Answer: C) $a^2 - 2ab + b^2$ **(1 M)**

19. 1 hectare = _____ square metres **[C]**

A) 100 B) 1,000 C) 10,000 D) 1,00,000

1 hectare = area of a square of side 100 m

= 100 × 100 = 10,000 m²

Final Answer: C) 10,000 **(1 M)**

20. In a rhombus ABCD, if ∠A = 50°, then ∠B = **[C]**

A) 50° B) 100° C) 130° D) 310°

Adjacent angles of a rhombus (parallelogram) are supplementary.

∠A + ∠B = 180°

∠B = 180° − 50° = 130°

Final Answer: C) 130° **(1 M)**

SECTION B – VERY SHORT ANSWER QUESTIONS (4 × 2 = 8 Marks)

21. Find the length of the side of a square whose area is 441 m².

Solution:

Step 1: Prime factorisation: 441 = 3 × 3 × 7 × 7 = $3^2 \times 7^2$ **(1 M)**

Step 2: Side = $\sqrt{441}$ = 3 × 7 = 21 m **(1 M)**

Final Answer: Side of the square = 21 m

Total: 2 Marks

22. Express 648 as a product of powers of its prime factors.

Solution:

Step 1: 648 = 2 × 324 = 2 × 2 × 162 = 2 × 2 × 2 × 81 **(1 M)**

Step 2: 81 = 3 × 3 × 3 × 3, so 648 = $2^3 \times 3^4$ **(1 M)**

Final Answer: 648 = $2^3 \times 3^4$

Total: 2 Marks

23. What will be the digital root of the number 9a + 36b + 13? Give a reason.

Solution:

Step 1: 9a and 36b are multiples of 9, so they leave remainder 0 when divided by 9 and do not change the digital root. **(1 M)**

Step 2: Digital root depends only on the remainder of 13 when divided by 9: 13 → 1 + 3 = 4. **(1 M)**

Final Answer: Digital root = 4

Total: 2 Marks

24. Solve the cryptarithm JK × 6 = KKK, where each letter stands for a digit and J ≠ 0.

Solution:

Step 1: JK = 10J + K and KKK = 111 × K. So 6(10J + K) = 111K, i.e. 60J + 6K = 111K. **(1 M)**

Step 2: 60J = 105K ⇒ 4J = 7K. The only digits satisfying this are J = 7, K = 4. Check: 74 × 6 = 444 ✓ **(1 M)**

Final Answer: J = 7, K = 4 (74 × 6 = 444)

Total: 2 Marks

SECTION C – SHORT ANSWER QUESTIONS (5 × 4 = 20 Marks)

25. Simplify and write the answers in exponential form:

(i) $10^{-2} \times 10^{-5}$ (ii) $5^7 \div 5^4$ (iii) $9^{-7} \div 9^4$ (iv) $(13^{-2})^{-3}$

Solution:

(i) $10^{-2} \times 10^{-5} = 10^{(-2)+(-5)} = 10^{-7}$ $[a^m \times a^n = a^{m+n}]$ **(1 M)**

(ii) $5^7 \div 5^4 = 5^{7-4} = 5^3$ $[a^m \div a^n = a^{m-n}]$ **(1 M)**

(iii) $9^{-7} \div 9^4 = 9^{-7-4} = 9^{-11}$ **(1 M)**

(iv) $(13^{-2})^{-3} = 13^{(-2)\times(-3)} = 13^6$ $[(a^m)^n = a^{mn}]$ **(1 M)**

Final Answer: (i) 10^{-7} (ii) 5^3 (iii) 9^{-11} (iv) 13^6

Total: 4 Marks

26. Write the following numbers in standard (scientific) form:

(i) 59,853 (ii) 65,950 (iii) 34,30,000 (iv) 70,04,00,00,000

Solution:

(i) 59,853 = 5.9853 × 10,000 = 5.9853 × 10^4 **(1 M)**

(ii) 65,950 = 6.595 × 10,000 = 6.595 × 10^4 **(1 M)**

(iii) $34,30,000 = 3.43 \times 10,00,000 = 3.43 \times 10^6$ **(1 M)**

(iv) $70,04,00,00,000 = 7.004 \times 10,00,00,00,000 = 7.004 \times 10^{10}$ **(1 M)**

Final Answer: (i) 5.9853×10^4 (ii) 6.595×10^4 (iii) 3.43×10^6 (iv) 7.004×10^{10}

Total: 4 Marks

27. Write the following numbers in the Roman numeral system:

(i) 1222 (ii) 2999 (iii) 302 (iv) 715

Solution: (Break each number into landmark numbers I = 1, V = 5, X = 10, L = 50, C = 100, D = 500, M = 1000)

(i) 1222 = 1000 + 100 + 100 + 10 + 10 + 1 + 1 = M CC XX II = MCCXXII **(1 M)**

(ii) 2999 = 1000 + 1000 + 900 + 90 + 9 = MM CM XC IX = MMCMXCIX **(1 M)**

(iii) 302 = 100 + 100 + 100 + 1 + 1 = CCC II = CCCII **(1 M)**

(iv) 715 = 500 + 100 + 100 + 10 + 5 = D CC X V = DCCXV **(1 M)**

Final Answer: (i) MCCXXII (ii) MMCMXCIX (iii) CCCII (iv) DCCXV

Total: 4 Marks

28. (i) Add the following Roman numerals and write the answer in Roman numerals: LXXXVII + LXXVIII **[2]**

(ii) Write the Hindu-Arabic numeral 25 in base-8 and in base-5. **[2]**

Solution:

(i) Count the symbols of both numbers together:

LXXXVII + LXXVIII has L : 2, X : 5, V : 2, I : 5 **(1 M)**

5 I = 1 V → V : 3, I : 0; 2 V = 1 X → X : 6, V : 1

5 X = 1 L → L : 3, X : 1; 2 L = 1 C → C : 1, L : 1

Result: C L X V = CLXV (check: 87 + 78 = 165 ✓)

(i) Answer: CLXV **(1 M)**

(ii) Base-8:

$25 = 3 \times 8 + 1$ (3 eights and 1 one) $\Rightarrow$ 25 = 31 in base 8 **(1 M)**

Base-5:

$25 = 1 \times 25 + 0 \times 5 + 0 \Rightarrow$ 25 = 100 in base 5 **(1 M)**

Final Answer: (i) CLXV (ii) 25 = 31 (base 8) = 100 (base 5)

Total: 4 Marks

29. If 48a23b is a multiple of 18, list all possible pairs of values for a and b.

Solution:

Step 1: A multiple of 18 is divisible by 2 and by 9. **(1 M)**

Step 2: Divisible by 2 $\Rightarrow$ last digit b is even: b = 0, 2, 4, 6 or 8. **(1 M)**

Step 3: Divisible by 9 $\Rightarrow$ digit sum 4 + 8 + a + 2 + 3 + b = 17 + a + b is a multiple of 9. Since $a + b \leq 18$, 17 + a + b = 18 or 27, so a + b = 1 or 10. **(1 M)**

Step 4: a + b = 1 with b even: (a, b) = (1, 0). a + b = 10 with b even: (2, 8), (4, 6), (6, 4), (8, 2). **(1 M)**

Final Answer: (a, b) = (1, 0), (2, 8), (4, 6), (6, 4), (8, 2)

Total: 4 Marks

SECTION D – LONG ANSWER QUESTIONS (4 × 8 = 32 Marks) (internal choice)

30. a) i) Find the smallest number by which 9408 must be multiplied so that the product is a perfect square. Also find the square root of the new number. **[4]**

ii) Find the cube roots of 27000 and 10648. **[4]**

(Or)

b) i) Find the smallest square number that is divisible by each of the numbers 4, 9 and 10. **[4]**

ii) What number should 1323 be multiplied by so that the product is a perfect cube? Also find the cube root of the new number. **[4]**

Solution – Option (a):

a) i) [4 Marks]

Step 1: Prime factorisation: $9408 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 3 \times 7 \times 7 = 2^6 \times 3 \times 7^2$ **(1 M)**

Step 2: Pair the prime factors: 2^6 and 7^2 form pairs; the factor 3 is left unpaired. **(1 M)**

Step 3: So the smallest multiplier is 3. New number = $9408 \times 3 = 28224 = 2^6 \times 3^2 \times 7^2$ **(1 M)**

Step 4: $\sqrt{28224} = 2^3 \times 3 \times 7 = 8 \times 21 = 168$ **(1 M)**

Final Answer: Smallest number = 3; square root of new number = 168

a) ii) [4 Marks]

Step 1: $27000 = 27 \times 1000 = 3^3 \times 10^3 = 2^3 \times 3^3 \times 5^3$ **(1 M)**

Step 2: $\sqrt[3]{27000} = 2 \times 3 \times 5 = 30$ **(1 M)**

Step 3: $10648 = 8 \times 1331 = 2^3 \times 11^3$ $(1331 = 11 \times 11 \times 11)$ **(1 M)**

Step 4: $\sqrt[3]{10648} = 2 \times 11 = 22$ **(1 M)**

Final Answer: $\sqrt[3]{27000} = 30$, $\sqrt[3]{10648} = 22$

Solution – Option (b):

b) i) [4 Marks]

Step 1: A number divisible by 4, 9 and 10 is divisible by their LCM. $4 = 2^2$, $9 = 3^2$, $10 = 2 \times 5$ **(1 M)**

Step 2: LCM $= 2^2 \times 3^2 \times 5 = 180$ **(1 M)**

Step 3: $180 = 2^2 \times 3^2 \times 5$ – the factor 5 is unpaired, so multiply by 5 to make a square. **(1 M)**

Step 4: $180 \times 5 = 900 = 2^2 \times 3^2 \times 5^2 = (2 \times 3 \times 5)^2 = 30^2$ **(1 M)**

Final Answer: Smallest square number = 900

b) ii) [4 Marks]

Step 1: $1323 = 3 \times 441 = 3 \times 3 \times 147 = 3 \times 3 \times 3 \times 49 = 3^3 \times 7^2$ **(1 M)**

Step 2: In a perfect cube every prime appears in groups of three; 7 appears only twice, so multiply by one more 7. **(1 M)**

Step 3: New number $= 1323 \times 7 = 9261 = 3^3 \times 7^3$ **(1 M)**

Step 4: $\sqrt[3]{9261} = 3 \times 7 = 21$ **(1 M)**

Final Answer: Multiply by 7; cube root of the new number = 21

Total: 8 Marks

31. a) i) Find all the other angles inside the rectangle ABCD in Figure (i). (The diagonals AC and BD meet at O and $\angle CAB = 30°$.) **[4]**

ii) Find all the other angles inside the rectangle PQRS in Figure (ii). (The diagonals PR and QS meet at O and $\angle QOR = 110°$.) **[4]**

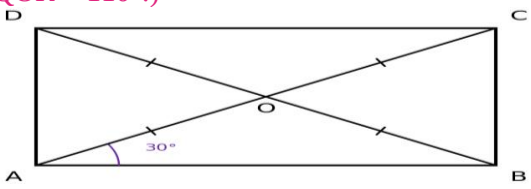


Figure (i)

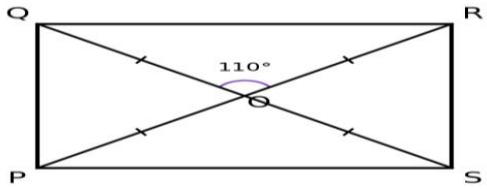
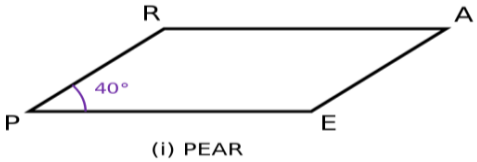


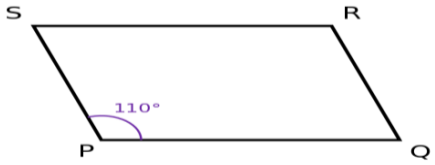
Figure (ii)

(Or)

b) i) Find the remaining angles in each of the following parallelograms (Figure (i) PEAR, $\angle P = 40°$; Figure (ii) PQRS, $\angle P = 110°$). **[4]**



(i) PEAR



(ii) PQRS

ii) Using the diagonal properties, construct a rhombus whose diagonals are of lengths 4 cm and 5 cm. (Write the steps of construction.) **[4]**

Solution – Option (a):

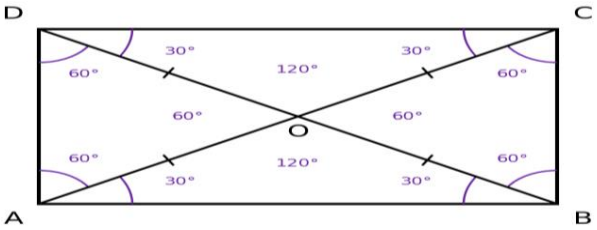
a) i) [4 Marks] (Figure (i): rectangle ABCD)

Step 1: Every angle of a rectangle is 90°: $\angle A = \angle B = \angle C = \angle D = 90°$. **(1 M)**

Step 2: $\angle DAC = \angle DAB - \angle CAB = 90° - 30° = 60°$. **(1 M)**

Step 3: Diagonals of a rectangle are equal and bisect each other, so OA = OB. In $\triangle OAB$, $\angle OBA = \angle OAB = 30°$, hence $\angle AOB = 180° - 30° - 30° = 120°$. Also $\angle OBC = 90° - 30° = 60°$. **(1 M)**

Step 4: $\angle BOC = 180° - 120° = 60°$; $\angle COD = \angle AOB = 120°$; $\angle AOD = \angle BOC = 60°$. AB ∥ DC gives $\angle ACD = \angle BDC = 30°$ (alternate angles); $\angle BCA = \angle ADB = 60°$. **(1 M)**



Final Answer: All angles are marked in the figure: 90° at each vertex; $\angle DAC = \angle BCA = \angle OBC = \angle ADB = 60°$; $\angle OBA = \angle ACD = \angle BDC = 30°$; $\angle AOB = \angle COD = 120°$; $\angle BOC = \angle AOD = 60°$.

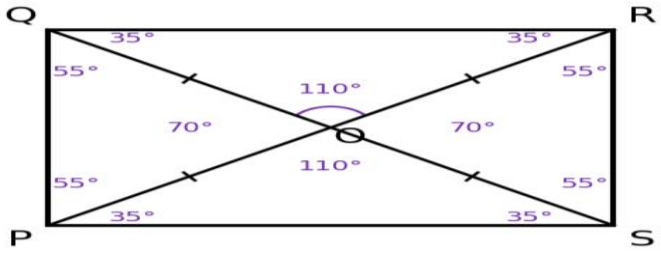
a) ii) [4 Marks] (Figure (ii): rectangle PQRS)

Step 1: All angles of the rectangle are 90°. PR is a straight line, so ∠POQ = 180° − ∠QOR = 180° − 110° = 70°. **(1 M)**

Step 2: Vertically opposite angles: ∠POS = ∠QOR = 110°, ∠ROS = ∠POQ = 70°. **(1 M)**

Step 3: OQ = OR, so in △OQR: ∠OQR = ∠ORQ = (180° − 110°) ÷ 2 = 35°. Similarly ∠OPS = ∠OSP = 35°. **(1 M)**

Step 4: ∠OQP = ∠QPO = (180° − 70°) ÷ 2 = 55° (check: 35° + 55° = 90°). Similarly ∠ORS = ∠OSR = 55°. **(1 M)**



Final Answer: ∠POQ = ∠ROS = 70°, ∠POS = 110°; ∠OQR = ∠ORQ = ∠OPS = ∠OSP = 35°; ∠OQP = ∠OPQ = ∠ORS = ∠OSR = 55°.

Solution – Option (b):

b) i) [4 Marks]

Parallelogram (i) PEAR with ∠P = 40°:

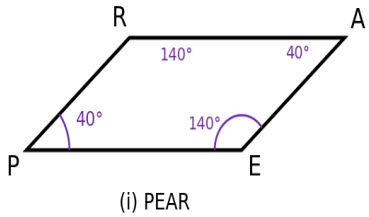
Step 1: Adjacent angles of a parallelogram are supplementary: ∠E = 180° − 40° = 140° and ∠R = 180° − 40° = 140°. **(1 M)**

Step 2: Opposite angles are equal: ∠A = ∠P = 40°. **(1 M)**

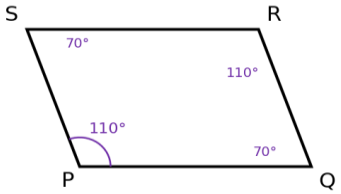
Parallelogram (ii) PQRS with ∠P = 110°:

Step 3: ∠Q = 180° − 110° = 70° and ∠S = 70° (adjacent angles supplementary). **(1 M)**

Step 4: ∠R = ∠P = 110° (opposite angles equal). **(1 M)**



(i) PEAR



(ii) PQRS

Final Answer: (i) ∠E = 140°, ∠A = 40°, ∠R = 140° (ii) ∠Q = 70°, ∠R = 110°, ∠S = 70°

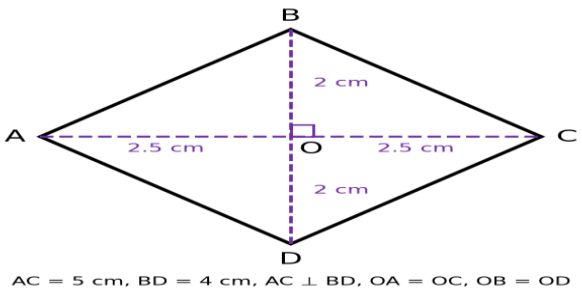
b) ii) [4 Marks] Construction of the rhombus

Step 1: Draw a line segment AC = 5 cm. **(1 M)**

Step 2: Draw the perpendicular bisector of AC; it meets AC at O (OA = OC = 2.5 cm). **(1 M)**

Step 3: On the perpendicular bisector mark points B and D on either side of O with OB = OD = 2 cm (BD = 4 cm). **(1 M)**

Step 4: Join AB, BC, CD and DA. ABCD is the required rhombus (diagonals bisect each other at 90°). **(1 M)**



AC = 5 cm, BD = 4 cm, AC ⊥ BD, OA = OC, OB = OD

Final Answer: ABCD is the required rhombus with diagonals AC = 5 cm and BD = 4 cm.

Total: 8 Marks

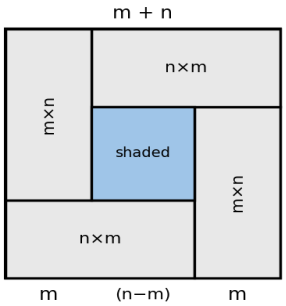
32. a) i) Find the following using the suggested identities: (a) 46^2 using $(a + b)^2$ (b) 397×403 using $(a + b)(a - b)$ (c) 91^2 using $(a - b)^2$ (d) 43×45 using $(a + b)(a - b)$ **[4]**

ii) Expand: $(a + b)(a^2 + 2ab + b^2)$ **[4]**

(Or)

b) i) Evaluate (using the fast-multiplication method): (a) 94×11 (b) 495×11 (c) 3279×11 (d) 89×101 **[4]**

ii) In the figure, a square of side $(m + n)$ is made of four identical rectangles of sides m and n, leaving a shaded square in the middle. Write the area of the shaded square (a) as $(m + n)^2 - 4mn$, and (b) as $(n - m)^2$. By expanding both expressions, show that $(m + n)^2 - 4mn = (n - m)^2$. **[4]**



Solution – Option (a):

a) i) [4 Marks]

(a) $46^2 = (40 + 6)^2 = 40^2 + 2 \times 40 \times 6 + 6^2 = 1600 + 480 + 36 = 2116$ **(1 M)**

(b) $397 \times 403 = (400 - 3)(400 + 3) = 400^2 - 3^2 = 160000 - 9 = 159991$ **(1 M)**

(c) $91^2 = (100 - 9)^2 = 100^2 - 2 \times 100 \times 9 + 9^2 = 10000 - 1800 + 81 = 8281$ **(1 M)**

(d) $43 \times 45 = (44 - 1)(44 + 1) = 44^2 - 1^2 = 1936 - 1 = 1935$ **(1 M)**

Final Answer: (a) 2116 (b) 159991 (c) 8281 (d) 1935

a) ii) [4 Marks]

Step 1: $(a + b)(a^2 + 2ab + b^2) = a(a^2 + 2ab + b^2) + b(a^2 + 2ab + b^2)$ [distributive property] **(1 M)**

Step 2: $= a^3 + 2a^2b + ab^2 + a^2b + 2ab^2 + b^3$ **(1 M)**

Step 3: Collect like terms: $2a^2b + a^2b = 3a^2b$ and $ab^2 + 2ab^2 = 3ab^2$ **(1 M)**

Step 4: $= a^3 + 3a^2b + 3ab^2 + b^3$ **(1 M)**

Final Answer: $(a + b)(a^2 + 2ab + b^2) = a^3 + 3a^2b + 3ab^2 + b^3$

Solution – Option (b):

b) i) [4 Marks]

(a) $94 \times 11 = 94 \times (10 + 1) = 940 + 94 = 1034$ **(1 M)**

(b) $495 \times 11 = 4950 + 495 = 5445$ **(1 M)**

(c) $3279 \times 11 = 32790 + 3279 = 36069$ **(1 M)**

(d) $89 \times 101 = 89 \times (100 + 1) = 8900 + 89 = 8989$ **(1 M)**

Final Answer: (a) 1034 (b) 5445 (c) 36069 (d) 8989

b) ii) [4 Marks]

Step 1: (a) Area of shaded square = area of big square − area of four rectangles = $(m + n)^2 - 4mn$. **(1 M)**

Step 2: (b) The shaded square has side $(n - m)$, so its area = $(n - m)^2$. **(1 M)**

Step 3: $(m + n)^2 - 4mn = m^2 + 2mn + n^2 - 4mn = m^2 - 2mn + n^2$ and $(n - m)^2 = n^2 - 2mn + m^2$. **(1 M)**

Step 4: Both expansions are equal to $m^2 - 2mn + n^2$. **(1 M)**

Final Answer: $(m + n)^2 - 4mn = (n - m)^2$ (proved)

Total: 8 Marks

33. a) i) (a) Divide ₹4,500 into two parts in the ratio 2 : 3. (b) In a science lab, acid and water are mixed in the ratio 1 : 5 to make a solution. How much acid and how much water does a bottle containing 240 mL of the solution have? **[4]**

ii) A mixture of 40 kg contains sand and cement in the ratio 3 : 1. How much cement should be added to the mixture to make the ratio of sand to cement 5 : 2? **[4]**

(Or)

b) i) Blue and yellow paints are mixed in the ratio 3 : 5 to produce green paint. How much of each colour is needed to produce 40 mL of green paint? To make a lighter shade, 20 mL of yellow is added to this mixture. What is the new ratio of blue to yellow? **[4]**

ii) A car travels 90 km in 150 minutes. If it continues at the same speed, what distance will it cover in 4 hours? **[4]**

Solution – Option (a):

a) i) [4 Marks]

(a) Total parts = 2 + 3 = 5; one part = ₹4,500 ÷ 5 = ₹900 **(1 M)**

First part = 2 × 900 = ₹1,800; second part = 3 × 900 = ₹2,700 **(1 M)**

(b) Total parts = 1 + 5 = 6; one part = 240 mL ÷ 6 = 40 mL **(1 M)**

Acid = 1 × 40 = 40 mL; water = 5 × 40 = 200 mL **(1 M)**

Final Answer: (a) ₹1,800 and ₹2,700 (b) Acid = 40 mL, Water = 200 mL

a) ii) [4 Marks]

Step 1: Sand : cement = 3 : 1, total 4 parts; one part = 40 ÷ 4 = 10 kg. Sand = 30 kg, cement = 10 kg. **(1 M)**

Step 2: Let x kg of cement be added. New ratio: 30 : (10 + x) = 5 : 2. **(1 M)**

Step 3: Cross-multiplying: $30 \times 2 = 5 \times (10 + x) \Rightarrow 60 = 50 + 5x \Rightarrow 5x = 10$ **(1 M)**

Step 4: $x = 2$. Check: $30 : 12 = 5 : 2$ ✓ **(1 M)**

Final Answer: 2 kg of cement should be added.

Solution – Option (b):

b) i) [4 Marks]

Step 1: Total parts = 3 + 5 = 8; one part = 40 mL ÷ 8 = 5 mL. **(1 M)**

Step 2: Blue = 3 × 5 = 15 mL; yellow = 5 × 5 = 25 mL. **(1 M)**

Step 3: After adding 20 mL yellow: yellow = 25 + 20 = 45 mL; blue = 15 mL. **(1 M)**

Step 4: New ratio blue : yellow = 15 : 45 = 1 : 3 (dividing by HCF 15). **(1 M)**

Final Answer: Blue = 15 mL, yellow = 25 mL; new ratio = 1 : 3

b) ii) [4 Marks]

Step 1: 4 hours = 4 × 60 = 240 minutes. **(1 M)**

Step 2: Time and distance are proportional: 150 : 90 :: 240 : x **(1 M)**

Step 3: Cross-multiplication: $150 \times x = 90 \times 240 \Rightarrow x = (90 \times 240) \div 150$ **(1 M)**

Step 4: $x = 21600 \div 150 = 144$ km **(1 M)**

Final Answer: Distance covered in 4 hours = 144 km

Total: 8 Marks



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